Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C++, SQL
Machine Learning: Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Anomaly Detection, Feature Engineering, Model Selection, Hyperparameter Tuning, Cross-Validation
Deep Learning: Neural Networks, CNNs, LSTM, Autoencoders, Federated Learning, Transfer Learning
Libraries & Frameworks: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV
Data Analysis & Visualization: EDA, Data Cleaning, Data Preprocessing, Statistical Analysis, Matplotlib, Seaborn, Power BI
Mathematics & Statistics: Probability, Descriptive Statistics, Hypothesis Testing, Linear Algebra, Calculus, Statistical Inference
Model Evaluation: Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, MAE, RMSE, Cross-Validation
Edge AI & Model Optimization: Model Compression, Quantization, TensorFlow Lite, TensorFlow Lite Micro (TFLM)
Databases & Tools: MySQL, SQLite, Git, GitHub, Jupyter Notebook, VS Code

EXPERIENCE

AI for Sustainability Virtual Intern **|LINK** — ***July – Sep, 2026***

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

Financial Transaction Anomaly Detection & Audit Dashboard **| LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

Customer Lifetime Value Engine **|LINK**

- Built BG/NBD and Gamma-Gamma probabilistic models on 1M+ transaction records to predict customer lifetime value and purchasing behaviour, incorporating frequency/recency analysis and model validation.
- Performed transaction-level preprocessing and value-based customer segmentation, identifying the top 20% of customers contributing ~65% of forecasted revenue.
- Developed Power BI dashboards to visualise customer segments, revenue distribution, and lifetime value predictions, enabling data-driven business decisions and customer prioritisation.

Real-Time Stock Market Anomaly Detector **|LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

V-FL: Privacy-Preserving Federated Learning for Edge IoT **|LINK**

- Developed a federated learning framework for privacy-preserving IoT anomaly detection using distributed model training.
- Implemented model compression and lightweight verification mechanisms to reduce communication overhead and detect model-update tampering.
- Evaluated robustness against malicious clients and model corruption, reporting approximately 19.7× compression and 100% tamper detection at 5% or greater corruption in your experiments.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** — 2025